Automatic Identification and Classification of Muscle Spasms in Long-Term EMG Recordings

Jeffrey Winslow, *Member, IEEE*, Adriana Martinez, and Christine K. Thomas

***Abstract*—Spinal cord injured (SCI) individuals may be afflicted by spasticity, a condition in which involuntary muscle spasms are common. EMG recordings can be analyzed to quantify this symptom of spasticity but manual identification and classification of spasms are time consuming. Here, an algorithm was created to find and classify spasm events automatically within 24-h recordings of EMG. The algorithm used expert rules and time-frequency techniques to classify spasm events as tonic, unit, or clonus spasms. A companion graphical user interface (GUI) program was also built to verify and correct the results of the automatic algorithm or manually defined events. Eight channel EMG recordings were made from seven different SCI subjects. The algorithm was able to correctly identify an average ($\pm$SD) of 94.5 $\pm$ 3.6% spasm events and correctly classify 91.6 $\pm$ 1.9% of spasm events, with an accuracy of 61.7 $\pm$ 16.2%. The accuracy improved to 85.5 $\pm$ 5.9% and the false positive rate decreased to 7.1 $\pm$ 7.3%, respectively, if noise events between spasms were removed. On average, the algorithm was more than 11 times faster than manual analysis. Use of both the algorithm and the GUI program provide a powerful tool for characterizing muscle spasms in 24-h EMG recordings, information which is important for clinical management of spasticity.**

***Index Terms*—Automatic classification, clonus, motor units, spasticity, spinal cord injury (SCI).**

I. Introduction

SPINAL CORD injured (SCI) individuals are commonly afflicted by spasticity, a complex phenomenon that may involve hyperreflexia, increased muscle tone resulting in higher resistance to muscle stretch at rest, and muscle spasms (involuntary muscle contractions), including the repeated contractions of clonus [1]. Studies on chronic SCI ($\geq$1 year post injury) subjects show that 65–75% of the sampled population have symptoms of spasticity. Spasticity can degrade the quality of life of individuals with SCI by restricting activities of daily living and rehabilitation, causing pain and fatigue, and by disturbing sleep [2]. Others use spasms to their advantage. For example, a SCI individual may know how to trigger a spasm and that contraction can shift his or her position in bed without any additional help.

Muscle spasms are common. They can occur throughout the day and night, and take various forms. The repetitive contractions of clonus can interfere with locomotion [3]. Sustained contractions can result in limb extension or flexion so make it hard to dress. Periodic limb movements (myoclonus) can disrupt sleep [4]. Additionally, trains of motor unit potentials can continue for minutes but may go unnoticed [5], [6].

Analysis of biological signals across time is one approach to understand complex phenomena. For example, long-term (24-h) electromyographic (EMG) recordings are useful to determine the characteristics of muscle spasms, particularly how common they are, and the prevalence of different types of contractions. In this way, EMG recordings can best provide a quantifiable measure to assess both some of the symptoms of spasticity and their severity. These long-term recordings generate large datasets, and multiple channels of EMG only compound the amount of data that must be processed. Presently, EMG experts can manually classify the different types of muscle spasms found in long-term recordings reliably [7], but the process is tedious and time consuming. These issues are not unique to EMG. Long-term ECG (electrocardiograms) and EEG (electroencephalograms) recordings also produce large quantities of data that are not feasible to analyze manually. Systems have been created to classify long-term ECG [8] and EEG [9], [10] automatically to look for biological anomalies in cardiac muscle and brain, respectively.

The aim of this paper was to produce an algorithm to automatically identify and classify muscle spasms in EMG records from different subjects and muscles by applying rules defined by experts in EMG recognition. The algorithm was designed to recognize spasm events, mark their start and end times, and appropriately label different contraction types. In addition, a graphical user interface (GUI) was created to verify spasms that were identified and classified manually by an expert with 5 years of experience in EMG recognition. The results of this manual (expert) spasm analysis were used to evaluate the performance of the automatic algorithm.

II. Methods

The subject pool consisted of seven SCI individuals (two women, five men; 27–52 years) who had a SCI at C4 ($n = 1$), C6 ($n = 5$) or C7 ($n = 1$) due to a diving ($n = 4$), motor vehicle ($n = 2$) or sports ($n = 1$) accident that occurred from 4 to 33 years ago. The American Spinal Injury Association Impairment scale was A ($n = 6$) or B ($n = 1$). All subjects signed a

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J. Winslow and A. Martinez are with The Miami Project to Cure Paralysis, University of Miami Miller School of Medicine, Miami, FL 33136 USA (e-mail: jwinslow@cableone.net; AMartinez9@med.miami.edu).

C. K. Thomas is with the The Miami Project to Cure Paralysis, Department of Neurological Surgery, Department of Physiology and Biophysics, University of Miami Miller School of Medicine, Miami, FL 33136 USA (e-mail: cthomas3@med.miami.edu).

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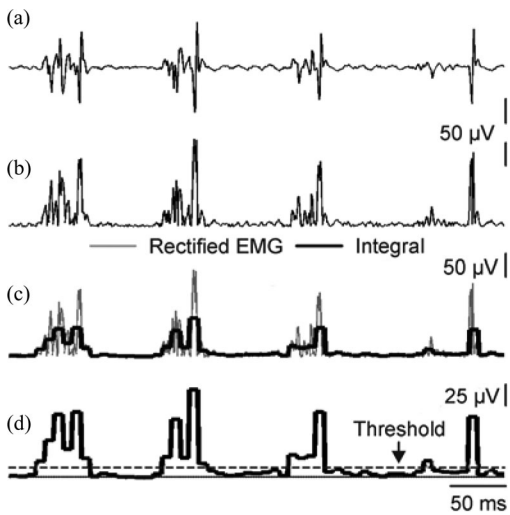

Fig. 1. EMG processing. EMG from Subject 1, Right medial gastrocnemius, 5 PM-6 PM (a) was rectified (b) and integrated (c) by calculating the area under the EMG every 10 ms. (d) Integrals greater than the threshold (dashed line) were considered EMG.

consent form approved by the University of Miami Investigational Review Board. One 24-h recording was made per subject.

A battery-powered data logging device with custom software [5] recorded EMG data from eight different leg muscles (right and left medial gastrocnemius, tibialis anterior, biceps femoris, vastus lateralis) for 24 h. All EMG signals were amplified on site (x400; Motion Lab Systems, Baton Rouge, LA, USA) and hardware filtered (30 Hz–1 kHz) before sampling at 1000 Hz. Each experiment consisted of 192 h of data per subject (8 channels × 24 h). EMG recording sites were as described by Mummidisetty *et al*. [16]. Briefly, self-adhering electrodes (Superior Silver, no. 626SS, Uni-Patch, Wabasha, MN, USA) were trimmed (1 cm × 3 cm) and positioned over the distal third of each muscle in a bipolar configuration (4-cm interelectrode distance). A ground electrode (1 cm × 3 cm) was positioned on each muscle, 4-cm proximal to the other two electrodes. The electrodes were held in place with Hypafix tape.

A. EMG Processing

The automatic algorithm used integrated EMG extensively to indicate muscle activity. The integral signal was constructed by rectifying the EMG and computing the area under the curve every 10 ms (see Fig. 1). The average duration of surface recorded motor unit potentials was 10 ms, and potentials from motor units were the minimum signal that was expected to be recorded [11]. To separate the EMG from the baseline, a threshold integral (mean of highest 10% of integrals + 3SD) was calculated for each channel of each recording from five 30-s regions of quiet baseline (no EMG activity; see [12]).

B. Rule-Based Identification and Classification

The automatic algorithm found all spasm events and then automatically classified them as tonic, unit, or clonus spasms.

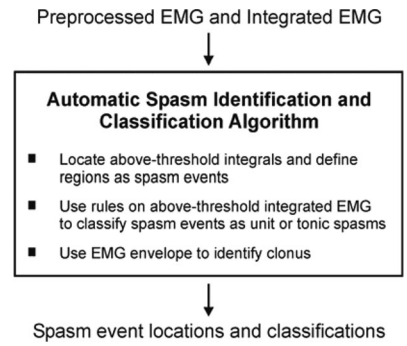

Fig. 2. Automatic algorithm flowchart. Spasm events were identified and classified based on integrated EMG above an integral threshold. The algorithm also identified clonus by investigating the frequency spectrum of the EMG envelope.

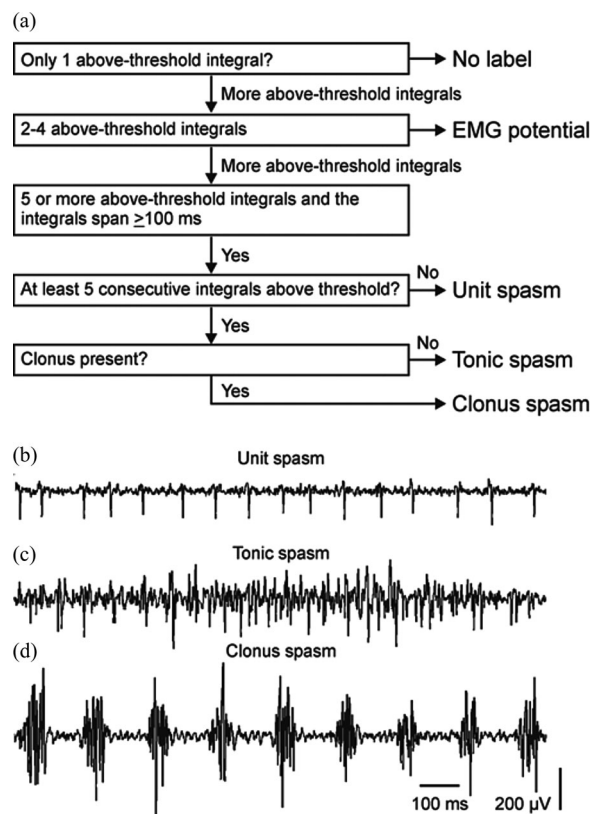

Fig. 3. Spasm classification rules flowchart. (a) Spasm events were classified according to how many consecutive above-threshold integrals they contained within a fixed time, and how many bursts of EMG were above threshold. Spasm events were classified as (b) unit, (c) tonic, or (d) clonus using rules described by Thomas *et al*. [7]. Unit spasms included distinct motor unit potentials, whereas motor unit potentials formed an interference pattern in tonic spasms. Clonus consisted of EMG bursts that occurred at frequencies between 4 and 12 Hz.

Fig. 2 details these steps of the algorithm. All processing was done one channel at a time.

Spasm events were identified using the same rules as experts [7]. Regions of EMG were grouped into segments such that spasm events were at least 1 s apart and had at least five integrals above threshold that spanned at least 100 ms.

Fig. 3(a) shows the rules used to classify spasm events [7]. An EMG region that had only two to four integrals above threshold

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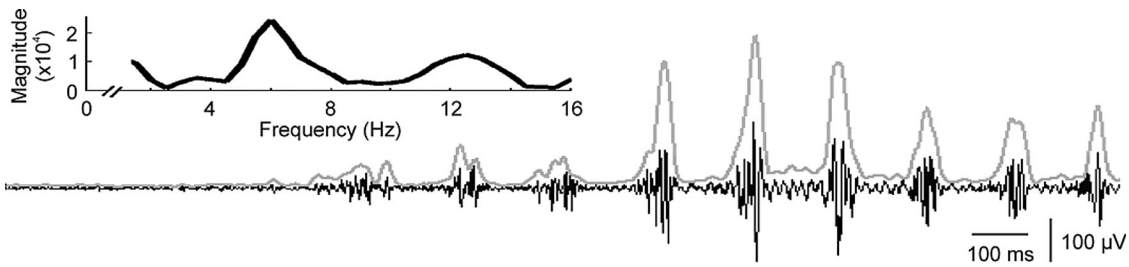

Fig. 4. Wavelet filtered EMG envelope and spectrogram (Subject 1, Right medial gastrocnemius, 8 AM–9 AM). EMG was wavelet filtered using a nonlinearly scaled Morlet wavelet (passband 74.8–193.9 Hz) to obtain an EMG envelope. The EMG envelope was subjected to a short-time Fourier transform (inset, left). A hamming window was applied to each EMG envelope window before taking the transform to minimize leakage of the spectral peaks. Two second windows ensured that at least four bursts of EMG (if they occurred at 4 Hz) were contained on either side of the window midpoint, which would meet the minimum number of EMG bursts necessary for clonus. Overlapping windows were used to enhance the time resolution of the frequency content. The primary clonus rhythm was at 6 Hz and this peak was greater than twice the next largest peak, so clonus was indicated.

in 100 ms was classified as an EMG potential. An event with 5 or more *nonconsecutive* above-threshold integrals was labeled a unit spasm as these data were potentials from one or more motor units [see Fig. 3(b)]. An event consisting of five or more *consecutive* integrals was labeled a tonic spasm because the EMG from various potentials formed an interference pattern [see Fig. 3(c)]. A tonic spasm was labeled clonus depending on the spectral analysis of the event EMG envelope (see Fig. 3(d); see clonus classification).

C. Clonus Classification

Clonus involves periodic bursts of EMG between 4 and 12 Hz, the frequency of which largely depend on the length of the reflex arc [13], [14]. Clonus is documented clinically when there are four bursts of EMG [15]. These two criteria were used here to indicate clonus (at least four EMG bursts, with periodicity between 4 and 12 Hz).

To identify clonus, wavelet, and short-time Fourier transforms were applied to all spasm events. First, an envelope of the EMG was created using a wavelet filtering technique based on nonlinearly scaling a mother Morlet wavelet. The resulting filter produced a pass band of 74.8–193.9 Hz [16]. For example, Fig. 4 shows a 2-s window of EMG that consists of more than four EMG bursts, the first criterion that had to be met to designate clonus. Next, a short-time Fourier transform was applied to the EMG envelope to locate the clonus periodicity using 2-s windows with 1.9-s overlaps. If the largest spectral peak within a 2-s window occurred at a frequency between 4 and 12 Hz, and that spectral peak was at least twice as large as the second largest spectral peak (which ensured the frequency component of the clonus dominated the spectrum), the second criterion for clonus was met. For example, the 6-Hz peak in Fig. 4 (inset, left) was 2.2 times the harmonic at 12 Hz, so clonus was deemed present.

III. Results

A. GUI to Assist Verification of Expert Spasm Analysis

A GUI program was created using MATLAB (The Mathworks, Natick, MA, USA) to assist the expert in verifying spasm identification and classification. The GUI program utilized MATLAB as an activeX server for DaDisp 2002 (DSP Development Corporation, Newton, MA, USA) so that

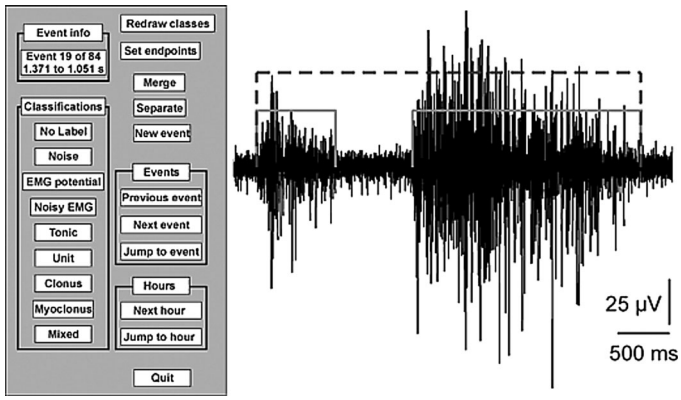

Fig. 5. MATLAB graphical user interface (GUI) program for viewing EMG. The GUI (left panel) executed commands to navigate through each hour of EMG, and enabled the user to change classifications or timings, create new events, merge, and separate existing spasm events. Different colored windows enclosed spasm events with different classifications. The merge tool within the GUI was used to combine (right panel) two contractions (solid and dashed lines) that were less than 1 s apart to adhere to spasm rules (Subject 1, Right vastus lateralis, 9 AM–10 AM).

commands executed by the GUI controlled the EMG data displayed by DaDisp 2002. Fig. 5 shows an example of the GUI control window (left) and EMG data (right) plotted with windows marking the start and end of spasms.

The GUI assisted the expert with 5 years of experience in EMG recognition in navigating through 24-h EMG recordings, changing spasm classifications or noise, and modifying event timing. The navigation buttons were used to either move to the previous event, advance to the next event, the next hour, or jump to an arbitrary hour. Classifications were changed by clicking the appropriate spasm type (left buttons). Existing events could be merged or separated. For example, the merge tool was used to join two manually labeled events that were less than 1 s apart (solid and dashed windows, right). New events could be created, in addition to modifying the start and end points of events. All changes made using the GUI were automatically saved and appended to the spasm event records for each recording.

B. Automatic Event Identification

First, the automatic algorithm was evaluated by comparing its ability to find spasm events to the results generated manually by

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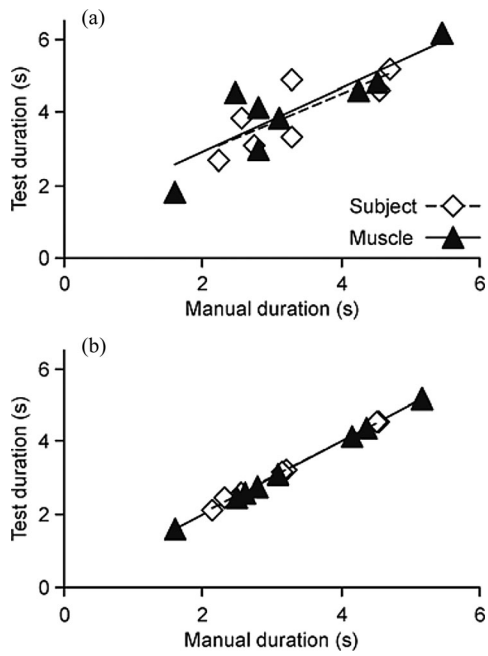

Fig. 6. Manual and test spasm durations. (A) Mean manual spasm durations plotted against automated (test) durations by subject (diamond, dashed line; $y = 0.8001x + 1.2619$, $R^2 = 0.658, p = 0.03$; least squares linear regression) and by muscle (triangle, solid line; $y = 0.8696x + 1.1484$, $R^2 = 0.748, p = 0.005$) and (B) after reduction in noise events (subject: $y = 0.9846x + 0.0755, R^2 = 0.998, p < 0.001$); muscle: $y = 1.02x - 0.0311$, $R^2 = 0.999, p < 0.001$).

an expert. Each event that was manually defined by an expert and verified using the GUI (hereafter referred to as a manual event) was compared with all automatically found events (test events). A test event was matched with a manual event if the durations of the test and manual events overlapped. If there was overlap only between one test event and one manual event, then that manual event was said to be a "one-to-one match." If multiple test events overlapped the same manual event, the result was a split (i.e., the manual event was split into multiple events by the automatic analysis). A combination occurred if multiple manually defined events were classified as part of one test event.

The automatic algorithm was able to identify an average ($\pm$SD) of 94.5 $\pm$ 3.6% (range: 89.3–98.8%) spasm events correctly over all seven experiments (5334 $\pm$ 2147 out of 5695 $\pm$ 2385 events per experiment were one-to-one matches). The unmatched events were either split into multiple events (0.7 $\pm$ 0.4%, 0–1.1%) or combined (4.8 $\pm$ 3.4%, 1.2–10%).

For one-to-one matches, there were significant positive correlations between the mean manual and test spasm durations by subject (diamonds) and by muscle (triangles), both before and after the number of noise events were removed (see Fig. 6). Only stand-alone noise, or noise that was not adjoining manual spasm events, was removed for these comparisons. The automatically found event durations tracked the manually found durations better after stand-alone noise events were removed [see Fig. 6(b)].

Before noise was removed, the average manual spasm duration by subject was 3.34 $\pm$ 0.96 s (range 2.74–4.57 s), which was significantly shorter than the corresponding automatically found duration (3.93 $\pm$ 0.94 s, 2.69–5.17 s, $p < 0.001$, two-way repeated measures ANOVA). Results also differed by muscle (manual: 3.08 $\pm$ 1.01 s, 1.61–5.47 s; automatic: 3.80 $\pm$ 1.07 s, 1.79–6.05 s, $p < 0.001$). When noise was removed, the average manual spasm duration per subject (3.20 $\pm$ 0.99 s, 2.15–4.55 s) was comparable to the automatically found duration (3.22 $\pm$ 0.98 s, 2.15–4.56 s, $p = 0.891$), as were durations by muscle (manual: 2.98 $\pm$ 0.93 s, 1.60–5.03 s; automatic: 3.01 $\pm$ 0.96 s, 1.60–5.07 s, $p = 0.891$).

C. Classification Performance Evaluation

The following definitions categorize the results of the automatic classification algorithm. A test event was a True Positive if its spasm type matched an overlapping manual event. A test event was a False Positive if no manual events overlapped it. A manual event was Missed if it did not overlap any test events. A test event was Misclassified if it overlapped a manual event with a different spasm type. If the class of a constituent test event of a split matched the spasm type of the manual event, that constituent event was a True Positive. All other test events that did not share the same spasm type as the manual event were False Positives. False positives were expressed as a percentage [test events that were incorrectly assigned a spasm class with respect to the total number of events (false positives, true positives) that were assigned that class (1)]. Accuracy for a given spasm type was the percentage of events that were correctly classified with respect to all spasm events that were assigned that spasm class

$$\%\text{FalsePostive} = N_{FP}/(N_{FP} + N_{TP}) \tag{1}$$

$$\%\text{Accuracy} = N_{TP}/(N_{FP} + N_{TP} + N_{\text{Misclassified}} + N_{\text{Missed}}). \tag{2}$$

In equations (1) and (2), N_{FP} is the number of false positive test events and N_{TP} is the number of true positive test events. In (2), $N_{\text{Misclassified}}$ is the number of misclassified test events, and N_{Missed} is the number of missed test events.

Fig. 7(a) shows the total number of true positive and misclassified spasm events by muscle. The percentage of true positives (range: 88–94% across muscles) and misclassified spasms (range: 6–12%) was comparable across muscles ($p = 0.882$; Chi square test). Overall, the automatic algorithm was able to correctly classify 91.6 $\pm$ 1.9% of all spasm events ($n = 7$ subjects, each experiment 24 h of data from eight muscles) with an accuracy of 61.7 $\pm$ 16.2%, and a false positive rate of 34.4 $\pm$ 17.8% (see Table I). The accuracy and false positive rate improved to 85.5 $\pm$ 5.9% and 7.1 $\pm$ 7.3%, respectively, if stand-alone noise events were removed. Out of 37,308 spasm events that were one-to-one matches, 2785 stand-alone noise events were removed (7.5%). Noise typically included high amplitude, transient spikes that were an order of magnitude larger than actual EMG, high-frequency spikes, or spikes that had broader durations than actual motor unit potentials. Some of the noise came from cell phones, electric wheelchairs, or mechanical artifacts due to uncontrolled limb movements. Experts could visually recognize these waveforms as noise versus EMG and manually excluded them from subsequent processing. Of all noise

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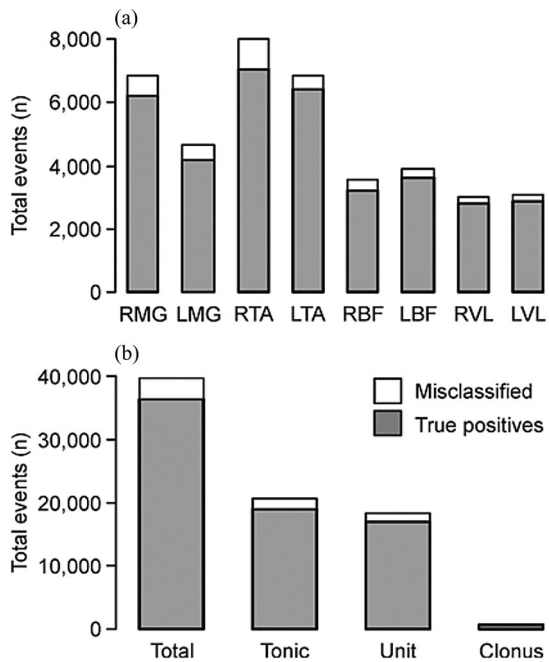

Fig. 7. Spasm events by muscle and type. Total number of true positive, and misclassified spasm events found by the automatic algorithm across muscles (a; $n = 7$ experiments) and by type (b).

TABLE I
OVERALL CLASSIFICATION PERFORMANCE

Statistic	Correct %	Accuracy %	Accuracy, Noise Removed %	False Positives %	False Positives, Noise Removed %
Mean	91.6	61.7	85.5	34.4	7.1
Max	94.5	86.1	91.7	58.4	21.8
Min	89.5	40.6	74.8	7.5	0.9
SD	1.9	16.2	5.9	17.8	7.3

TABLE II
ALGORITHM PERFORMANCES BY SPASM TYPE

Spasm Type	Accuracy % Mean ± SD	Accuracy, Noise Removed % Mean ± SD
Tonic	65.3 ± 15.2	85.7 ± 9.4
Unit	58.6 ± 18.4	87.5 ± 6.2
Clonus	42.9 ± 17.3	67.9 ± 19.9

events, 80% had durations less than 3.2 s, the average duration of spasms.

Fig. 7(b) shows the total number of true positive and misclassified spasm events with respect to spasm type. Across all subjects, most spasms were either tonic (52%) or unit spasms (46%). Only 2% of spasms were labeled as clonus. Of these clonus spasms, 67% were true positives. The percentage of true positives for tonic (91%) and unit (92%) spasms was comparable (p $= 0.79$) but lower for clonus ($p < 0.001$; Chi square tests). Overall, tonic and unit spasms were classified to an accuracy of 85.7 $\pm$ 9.4% and 87.5 $\pm$ 6.2% when stand-alone noise was removed, respectively (Table II, $n = 7$ experiments). The automatic algorithm performed poorest on clonus, achieving an accuracy of 67.9 $\pm$ 19.9% after noise removal.

TABLE III
EXECUTION TIMES

Processing Type	Per Hour	Per Experiment (8 Channels)
Manual	39.8 ± 26.0 min (2389 s)	127 Hrs, 56 min
Algorithm	3.4 ± 0.5 min (206 s)	10 Hrs, 58 min
GUI-assisted	10.8 ± 4.1 min (650 s)	34 Hrs, 39 min

D. Execution Times

Since the algorithm cannot automatically identify noise, its results must be verified by an expert (i.e., GUI-assisted analysis). Thus, to evaluate the execution time we conducted a three-way comparison of analysis methods (manual versus automatic versus GUI-assisted) for 15 h of data with average amounts of EMG. Execution time was measured using the computer clock. The average time required for an expert to manually analyze and verify 1 h of data (Intel Core i7 machine, 24 GB RAM) was 39.8 $\pm$ 26.0 min (using DaDisp cursors, mouse clicks, and keyboard navigation), significantly longer than the automatic algorithm (3.4 $\pm$ 0.5 min; $p = 0.017$; Intel Core i5 machine, 8 GB RAM) or the GUI-assisted method (Intel Core i7 machine) which involved running the automatic algorithm and the time required to verify these results using the GUI (10.8 $\pm$ 4.1 min; $p = 0.025$; Repeated measures ANOVA; Table III). When these results were extrapolated to the time it would take to analyze an entire experiment (192 h of data), automatic classification using the algorithm was 11 times faster than manual analysis. GUI-assisted analysis was almost four times faster than manual analysis.

IV. DISCUSSION

An algorithm was created to automatically find muscle spasms in long-term EMG recordings and to classify the spasm type. The algorithm employed the same timing and classification rules developed by experts in EMG recognition, removing subjective judgment calls. A GUI was also created to allow effortless navigation through the long EMG recordings so an expert could verify or modify spasm events and classifications generated automatically or manually. Not only was the automated method significantly quicker than manual analysis, verification with the GUI increased classification accuracy to match that of an expert. The automatic spasm identification, classification, and viewing system developed here provide a powerful tool to quantify and assess the involuntary muscle contractions that are characteristic of spasticity.

A. Accurate and Automatic Identification of Muscle Spasms

An algorithm can only be expected to perform as well as the agreement between two experts in EMG recognition, because individuals can make different subjective judgments when utilizing the same rules. Elsewhere, we have systematically assessed

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the interrater reliability of our spasm identification and classification 14]. Two experts evaluated 48 h of EMG data using the same rules that the algorithm used. These results are discussed concurrently with the automatic algorithm results. The algorithm developed here correctly identified 94.5 $\pm$ 3.6% of all spasm events as one-to-one matches because it employed the same above-threshold integral, 1 s, and 100 ms timing rules developed by experts. In contrast, experts missed 1.8% of spasms using these same rules. Experts marked another 1.9% of events that did not meet spasm criteria. Thus, an algorithm is much better suited to counting above-threshold integrals than an expert when there are a large number of spasms to be found in multichannel EMG recordings. Operator fatigue can set in and the actual number of above threshold integrals can be miscounted.

Experts combined an average of 4.7% spasm events which was similar to the 4.8% of spasm events that the algorithm combined. Expert combinations resulted from grouping various contractions as myoclonus or from misapplying rules, such as the 1-s rule, whereas the algorithm combinations were due to noise occurring within or between different spasm events. Noise within spasm events was excluded and labeled during manual classification, but not by the classification algorithm, leading to combinations and longer duration spasms. Automatically identifying noise was not a goal of this study, but will be addressed in future work through software and hardware measures. Nevertheless, when noise was included, the algorithm correctly deduced the spasm start and end times within an accuracy of 1 ms (one sampling period) in 89% and 90% of all one-to-one matched events, respectively. When stand-alone noise was removed, the average spasm length found manually by subject (3.20 $\pm$ 0.99 s) and muscle (2.98 $\pm$ 0.93 s) was on par with the average automatically found durations (subject: 3.22 $\pm$ 0.98 s; muscle: 3.01 $\pm$ 0.96 s).

B. Different Spasm Types Can be Classified Automatically

The algorithm showed good classification performance. A large percentage (91.6 $\pm$ 1.9%) of spasms was classified correctly, both across muscles and subjects (see Fig. 7). This was due to the fact that unit and tonic spasms could be appropriately identified 92% and 91% of the time, respectively, using the simple rule of either five nonconsecutive or consecutive integrals that exceeded the threshold [see Fig. 7(b)]. Misclassifications largely occurred due to noise that either stood-alone or was mixed together with actual spasm events. For example, some unit spasms were misclassified as tonic spasms because the algorithm identified above-threshold noise as EMG. An expert would label noise separately, whereas the algorithm was not programmed to do so. After the stand-alone noise was removed, algorithm accuracy averaged 85.5 $\pm$ 5.9%. Two raters agreed on spasm classification 89% of the time [7], results that were comparable to the algorithm ($p = 0.669$, Chi square test).

Clonus was difficult to classify. When noise was removed, the accuracy of identifying clonus averaged 67.9 $\pm$ 19.9% (see Table II). For simplicity, clonus was identified by inspecting the frequency content of the EMG envelope using a short-time Fourier transform. Better performance may be obtained by modifying the clonus finding portion of the automatic algorithm to identify the harmonics at multiples of the base clonus frequency (see Fig. 4) instead of just relying on spectrum peak ratios. Additionally, wavelet transforms could be used to analyze the EMG envelope signal because they provide better time resolution than a short-time Fourier transform [16], [17]. Even though clonus had been previously analyzed using wavelets, it was not automatically classified as accomplished by the current algorithm. Mummidisetty *et al.* [16] identified EMG burst start and end times within already recognized clonus.

The timeframe for identifying and classifying muscle spasms from EMG recordings has different requirements from those of other signals like EEG. When EEG data are monitored acutely for the presence of spikes that indicate seizures, high accuracy is required immediately in order to make clinical decisions [9], [10]. In contrast, spasm identification and classification can be done offline. Longer EMG datasets are required to build a global view of muscle activity over time so that spasms with functional consequences can be identified and managed clinically.

C. Use of a GUI Can Improve the Accuracy of Automatic Spasm Analysis

A GUI was created to enable human operators to easily navigate through long EMG recordings using a point and click interface, and to verify or modify manual or automatic spasm analysis where appropriate. Using the GUI only warranted an additional 7.4 $\pm$ 3.7 min to verify the automatic classifications of each hour of EMG, which is almost five times less manual work than classifying that same hour by hand (see Table III). This verification process increased analysis accuracy to match manual levels.

The flexibility of the GUI is a strength. There are multiple approaches that can be used to fix a given event. For example, if only the last part of an event is noise, it could be divided in two and each part reclassified. Another approach would be to change the start and end of the noise, then to create a new event for the EMG.

Other approaches could also be implemented to improve performance. If the automatic classification algorithm only found and classified events as unit or tonic spasms, execution times would be twice as fast (103 s to process 1 h of data compared to 206 s) because spectrogram calculations to identify clonus were computationally intensive. This would yield execution times of 5 h, 29 min per 8-channel experiment which could be accomplished overnight. Across seven subjects, clonus made up only 2% of the total number of spasm events. The number of spasms involving myoclonus was also low and only its constituent events were marked as tonic, unit or clonus spasms here. However, if the algorithm was used with the GUI program developed here, an expert would only need to change a few classifications (mainly clonus and myoclonus) because unit and tonic spasms were classified correctly to a high degree (91.9 $\pm$ 3.5% and 90.7 $\pm$ 3.1%, respectively). This would speed up the verification process.

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D. Automatic Analysis of Muscle Spasms Allows Important Rehabilitation Questions to be Addressed Comprehensively Using Large Datasets

The combination of automated analysis and computer-assisted verification of results by EMG experts enhanced the accuracy of muscle spasm analysis in 24-h EMG records and reduced the analysis time well below that required to accomplish this work manually. The algorithm results are comparable to the performance of two experts with respect to identification of spasm events (95% versus 96% one-to-one matches found), but automated analysis was 11 times faster than manual analysis. Automatically, generated spasm classifications were correct 92% of the time, on average, with an accuracy of 86% after noise removal. These results compare favorably to the classification agreement found between two experts in EMG recognition (89%). Thus, the automated analysis performs just as well as two experts, but the execution time is much faster and less tedious.

Analysis of an entire day of EMG obtained from multiple muscles in everyday settings allows some of the most problematic symptoms of spasticity to be quantified objectively during both wake and sleep time. When linked to the individual's perception of these muscle spasms, these data will provide insight into how these involuntary contractions interfere with rehabilitation, care, mobility, and sleep. Clinical assessments rely on subjective spasm counts but contraction duration, intensity, type, or the particular combination of active muscles may also make spasms disruptive. Long-term EMG records provide this information and can be used to ask clinically important questions about involuntary muscle contractions. For example, examination of the likelihood that a certain type of spasm may occur again, or how spasms are influenced by medication or training, may lead to improved clinical management of these contractions.

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Jeffrey Winslow (M'95) received the Ph.D. degree from the University of Miami, FL, USA, in 2008.

He is currently a Professor at Yavapai College, Prescott, AZ, USA. His research interests include muscle spasticity and the processing of biological signals.

Adriana Martinez received the B.Sc. degree in mathematics from Havana University, Havana, Cuba.

She is currently a graduate student in Statistics at George Mason University, Fairfax, VA and a Research Support Specialist at The Miami Project to Cure Paralysis, Miami, FL.

Christine K. Thomas received the Ph.D. degree from the University of Alberta, Canada, in 1986.

She is a Professor at The Miami Project to Cure Paralysis, at the University of Miami, FL, USA. Her research examines the consequences of spinal cord injury on skeletal muscle, including muscle atrophy, weakness, denervation, fatigue, and spasticity.

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